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Subharmonic Solutions of a Pendulum Under Vertical Anharmonic Oscillations of the Point of Suspension

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Abstract—We prove the existence of subharmonic solutions in the dynamics of a pendulum whose point of suspension executes a vertical anharmonic oscillation of small amplitude.

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To Prof. A. P. Markeev, on his 75th anniversary

1. INTRODUCTION

We consider the motion of a planar mathematical pendulum with the point of suspension performing an anharmonic vertical oscillation of small amplitude. The same problem with the point of suspension performing a harmonic vertical oscillation was considered in [10] from the viewpoint of stability of the equilibrium positions. In [13] the existence of a family of periodic solutions in a small neighborhood of the stable equilibrium of the unperturbed pendulum is proved for the case of a horizontal harmonic oscillation of the suspension point. This family, of course, contains many subharmonic solutions, but they are all near the equilibrium. Stability of these periodic solutions was also considered in that work. In [14] Leung and Kuang study the chaotic dynamics of a spherical pendulum with a harmonically vibrating suspension point. Here we consider the dynamics from the point of view of periodic orbits.

Our system, see Eq. (2.6), is a periodic perturbation of the pendulum with a fixed suspension point. In the phase space of the unperturbed pendulum a region of oscillations, that is, a region bounded by the separatrices of the pendulum with a fixed suspension point is filled with periodic solutions $\mathbf{x}_k(\tau)$, $0 \leqslant k < 1$, of this pendulum whose period T_k is a differentiable function of k and increases from the period of small oscillations to infinite as k varies from 0 to 1 and the derivative of T_k with respect to k is positive for all k. The motion of the suspension point is taken as a vertical anharmonic oscillation parameterized by a multiple of the time in such a way that the period of the perturbation is independent of the small parameter ϵ, a quantity proportional to the amplitude of the oscillation of the suspension point, see Section 2. The subharmonic method of Melnikov for searching for periodic solutions of perturbations of planar Hamiltonian systems is a technique that may be applied in this context, see Section 3. Our goal is to prove the following theorem.

Theorem 1. *Let $\mathbf{x}_k(\tau)$ be a solution of the unperturbed system in (2.6) with period $T_k = \frac{m}{n}T_0$ where T_0 is the period of the perturbation and m, n are relatively prime positive integers. Then there is an $\epsilon_n > 0$ such that for all $\epsilon \leqslant \epsilon_n$ there is a solution $\mathbf{x}(\tau, \epsilon)$ of the perturbed system with period $nT_k = mT_0$ which stays close to $\mathbf{x}_k(\tau)$ for all time.*

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2. THE EQUATION OF MOTION

Consider a pendulum moving in a vertical plane with a suspension point oscillating vertically. Take a fixed coordinate system Oxy with the origin on the vertical line containing the suspension point, axis Ox directed vertically downwards and axis Oy horizontally oriented to the right. Let $\mathbf{r} = OP$ be the position vector of a mass m placed at the point P, free extremity of the rod of the pendulum, of length l. Let $\rho = \rho(1, 0)$ be the position vector of the suspension point. The vector $(1, 0)$ is directed downwards and the vector $(0, 1)$ to the right. Let θ be the angle that the rod makes with the vector $(1, 0)$. Let $\mathbf{e}_1 = (\cos\theta, \sin\theta)$ and $\mathbf{e}_2 = (-\sin\theta, \cos\theta)$, so $\dot{\mathbf{e}}_1 = \dot{\theta}\mathbf{e}_2$ and $\dot{\mathbf{e}}_2 = -\dot{\theta}\mathbf{e}_1$.

Since

$$\mathbf{r} = \rho + l\mathbf{e}_1,$$

differentiating with respect to time t, we have

$$\ddot{\mathbf{r}} = (\ddot{\rho}\cos\theta - l\dot{\theta}^2)\mathbf{e}_1 + (l\ddot{\theta} - \ddot{\rho}\sin\theta)\mathbf{e}_2.$$

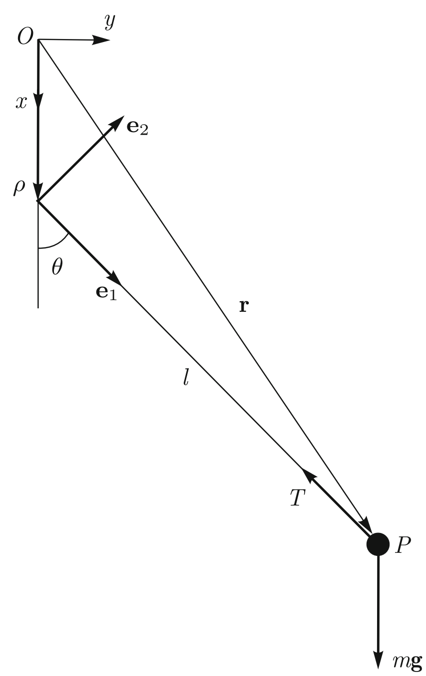

Fig. 1. Pendulum with vertical oscillation of the point of suspension.

The force $\mathbf{F}$ at point P is the sum of the weight of mass m with the traction of the rod. Since $(1, 0) = \cos\theta\mathbf{e}_1 - \sin\theta\mathbf{e}_2$, we have the following expression for it:

$$\mathbf{F} = (mg\cos\theta - T)\mathbf{e}_1 - mg\sin\theta\mathbf{e}_2.$$

Comparing the components of the vector $\mathbf{e}_2$ in the equation given by the second law of dynamics, $m\ddot{\mathbf{r}} = \mathbf{F}$, we obtain the equality $l\ddot{\theta} - \ddot{\rho}\sin\theta = -g\sin\theta$, and therefore the equation of motion of mass m is

$$\ddot{\theta} + \frac{g}{l}\sin\theta - \frac{\ddot{\rho}}{l}\sin\theta = 0. \tag{2.1}$$

The components of the vector $\mathbf{e}_1$ give the traction T at each instant of time.

Consider ρ to be the solution $\rho(t) = a\cos\omega t$ of the harmonic oscillator with frequency $\omega = \sqrt{k_1}$ and amplitude a. Taking $\tau = \sqrt{k_1}\,t$ as the new time and denoting by $'$ derivatives with respect to τ, we have

$$\theta'' + \frac{g}{k_1 l}\sin\theta - \frac{\rho''}{l}\sin\theta = 0. \tag{2.2}$$

Since $\rho'' = -a\cos\tau$, setting $\alpha = \frac{g}{k_1 l}$, $\beta = \frac{a}{l}$, the equation of motion (2.1) takes the form

$$\theta'' + (\alpha + \beta\cos\tau)\sin\theta = 0,$$

which is the equation of motion of the pendulum under a vertical harmonic oscillation of the suspension point, see [10].

We now consider the case of an anharmonic oscillation of the point of suspension. Let $\rho_3 < 0 < \rho_2 < \rho_1$ be the roots of the equation (see Fig. 2)

$$2h - k_1\tilde{\rho}^2 - \frac{1}{3}k_2\tilde{\rho}^3 = 0, \quad h > 0,\ k_1 > 0,\ k_2 < 0$$

and consider the differential equation of the anharmonic oscillator with time ζ

$$\frac{d^2\tilde{\rho}}{d\zeta^2} = -k_1\tilde{\rho} - \frac{1}{2}k_2\tilde{\rho}^2.$$

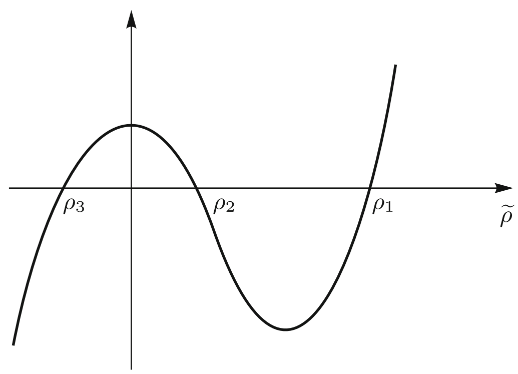

Fig. 2. Roots of the cubic polynomial.

Its solutions are given by (see [8])

$$\tilde{\rho}(\zeta) = \rho_3 + (\rho_2 - \rho_3)\mathrm{sn}^2(\sigma, k_0), \tag{2.3}$$

where $\mathrm{sn}(\sigma, k_0)$ is the Jacobi elliptic sine with module k_0,

$$k_0^2 = \frac{\rho_2 - \rho_3}{\rho_1 - \rho_3}, \quad \sigma = \sqrt{\alpha(\rho_1 - \rho_3)}\,\zeta \quad \text{and} \quad 4\alpha = -\frac{k_2}{3}.$$

Take $\tau = \sqrt{k_1}\,t$ so that Eq. (2.1) is transformed into Eq. (2.2).

Let $K_0 = K(k_0)$ be the complete elliptic integral of first kind with module k_0, take

$$\zeta = \frac{K_0}{\pi\sqrt{\alpha(\rho_1 - \rho_3)}}\,\tau,$$

and consider for $\rho(\tau)$ in (2.2) the function defined by $\rho(\tau) = \tilde{\rho}(\zeta)$. Since $\frac{d\sigma}{d\zeta}\frac{d\zeta}{d\tau} = \frac{K_0}{\pi}$, we have by the chain rule

$$\rho' = \frac{d\rho}{d\tau} = \frac{d\tilde{\rho}}{d\sigma}\frac{d\sigma}{d\zeta}\frac{d\zeta}{d\tau} = \frac{K_0}{\pi}\frac{d\tilde{\rho}}{d\sigma}, \qquad \rho'' = \left(\frac{K_0}{\pi}\right)^2\frac{d^2\tilde{\rho}}{d\sigma^2}.$$

From (2.3) we have $\frac{d\tilde{\rho}}{d\sigma} = 2a\,\mathrm{sn}(\sigma, k_0)$, where $a = \rho_2 - \rho_3$, so $\frac{d^2\tilde{\rho}}{d\sigma^2} = 2a\Sigma(\sigma)$, with

$$\Sigma(\sigma) = \mathrm{cn}^2(\sigma, k_0)\mathrm{dn}^2(\sigma, k_0) - \mathrm{sn}^2(\sigma, k_0)\mathrm{dn}^2(\sigma, k_0) - k_0^2\,\mathrm{sn}^2(\sigma, k_0)\mathrm{cn}^2(\sigma, k_0). \tag{2.4}$$

The functions $\mathrm{sn} = \mathrm{sn}(\sigma, k_0)$, cn and dn are the Jacobi elliptic functions with module k_0.

Therefore, $\rho'' = 2a\left(\frac{K_0}{\pi}\right)^2 \Sigma(\sigma)$, so finally, setting $Z(\tau) = \Sigma(\sigma)$, Eq. (2.2) is written in the form

$$\theta'' + \omega^2 \sin\theta - \epsilon Z(\tau)\sin\theta = 0, \tag{2.5}$$

where

$$\omega^2 = g/k_1 l \qquad \text{and} \qquad \epsilon = \frac{2a}{l}\left(\frac{K_0}{\pi}\right)^2.$$

We will take $k_0 \leqslant 1/2$, so ϵ is a small parameter that goes to zero as k_0 goes to zero.

As $\sigma = \frac{K_0}{\pi}\tau$, we have

$$Z(\tau + 2\pi) = \Sigma\left(\frac{K_0}{\pi}(\tau + 2\pi)\right) = \Sigma\left(\frac{K_0}{\pi}\tau + 2K_0\right) = Z(\tau).$$

Notice that the period $T_0 = 2\pi$ does not depend on ϵ. Taking $x = \theta$ and $y = \theta'$, the equation of motion (2.5) is written as the system in the plane

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} y \\ -\omega^2\ \sin x \end{bmatrix} + \epsilon \begin{bmatrix} 0 \\ Z(\tau)\sin x \end{bmatrix}. \tag{2.6}$$

3. A METHOD FOR SEARCHING SUBHARMONIC SOLUTIONS

The system of Eqs. (2.6) is a time-periodic perturbation of the system describing the motion of a pendulum with a fixed suspension point. The unperturbed system is an autonomous one-degree-of-freedom Hamiltonian system. Its phase portrait is shown on the right. In a region bounded by the separatrices (a region of oscillations) the solutions are all periodic with the orbits emanating from the stable equilibrium and filling this region up to the boundary curves, the separatrices. The period of these solutions increases from the period of small oscillations to infinity.

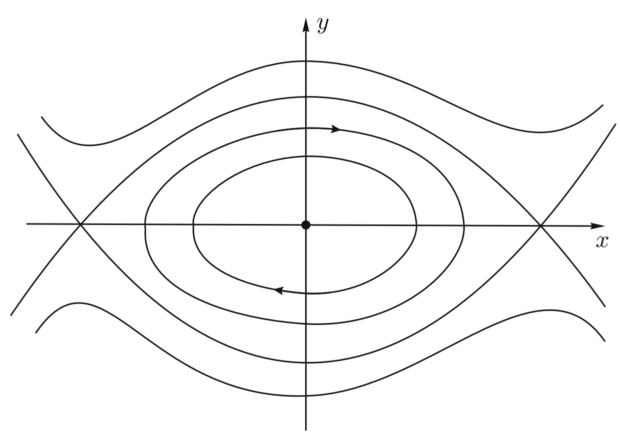

Fig. 3. Phase portrait of the pendulum.

The question arises whether from one of the periodic solutions of the unperturbed system with period T we can prove the existence of a periodic solution of the unperturbed system (2.6) with a period multiple of T. One technique to approach this problem is the subharmonic Poincaré–Melnikov method, see Section 46, Chapter 3 of [6] and Part 2 of [15].

The above-mentioned method is the following. Consider a planar system

$$\dot{\mathbf{x}} = f(\mathbf{x}) + \epsilon g(\mathbf{x}, t), \qquad \mathbf{x} \in \mathbf{R}^2, \tag{3.1}$$

where the unperturbed system $\dot{\mathbf{x}} = f(\mathbf{x})$ is Hamiltonian and the perturbation $g(\mathbf{x}, t)$ is periodic in t with period T_0, independent of ϵ; it does not need to be Hamiltonian. It is assumed that the unperturbed system has a homoclinic loop that bounds a region filled with periodic orbits $\mathbf{x}_\mu(t)$ that emanate from a stable equilibrium and whose period is a differentiable function of the parameter μ having a positive derivative with respect to μ. For μ we can take any parameter that uniquely determines each closed orbit of the unperturbed system. Two separatrices that bound a region as in the case of the pendulum serve the same purpose as a homoclinic loop. Let $\mathbf{x}_\mu(t)$ be a periodic solution of the unperturbed system with period $T_\mu = \frac{m}{n}T_0$, where m and n are relatively prime positive integers.

Now consider the function defined by

$$M_\mu(t_0) = \int_0^{nT_\mu} f\big(\mathbf{x}_\mu(t + t_0)\big) \wedge g\big(\mathbf{x}_\mu(t + t_0), t\big)\, dt, \tag{3.2}$$

where $(a_1, a_2) \wedge (b_1, b_2) = a_1 b_2 - a_2 b_1$. The Poincaré–Melnikov theorem says that if $M_\mu(t_0)$ has a simple zero, then there is an $\epsilon_n > 0$ such that for $0 < \epsilon \leqslant \epsilon_n$ there is a periodic solution of the perturbed system (3.1) with a period equal to mT_0. A proof of this theorem can also be found in [11] or [3].

4. THE GENERATING FUNCTION FOR SUBHARMONIC SOLUTIONS

The solution of the unperturbed system in (2.6), that is, the pendulum, $x'' + \omega^2 \sin x = 0$, with the initial condition $x(0) = 0$, $y(0) = 2k\omega$, is given by (see any text in mechanics, for instance, Appendix 2 of [4] or Chapter XI of [5])

$$x(\tau) = 2\mathrm{arcsin}\big(k\,\mathrm{sn}(\omega\tau, k)\big), \qquad y(\tau) = 2k\omega\mathrm{cn}(\omega\tau, k),$$

where $\mathrm{sn}(\omega\tau, k)$ and $\mathrm{cn}(\omega\tau, k)$ are the Jacobi elliptic sine and cosine of module k. This solution is periodic with period $T_k = 4K/\omega$, where $K = K(k)$ is the complete elliptic integral of first kind with module k.

Now $\sin(2\mathrm{arcsin}\xi) = 2\sin(\mathrm{arcsin}\xi)\cos(\mathrm{arcsin}\xi) = 2\xi\sqrt{1-\xi^2}$, hence for $\xi = k\,\mathrm{sn}\,\omega\tau$ we obtain

$$\sin x(\tau) = 2k\,\mathrm{sn}\,\omega\tau\,\mathrm{dn}\,\omega\tau.$$

In the quality of the parameter μ of the previous section we take the value $\mu = k$ of the module of the elliptic function $\mathrm{sn}(\omega\tau, k)$; so $0 \leqslant k < 1$.

For the integrand of the function $M_k(\tau_0)$ in (3.2), we have from (2.6) $f \wedge g = Z(\tau)\; y(\tau + \tau_0)\sin x(\tau + \tau_0)$, and therefore we get

$$f \wedge g = 4k^2\omega Z(\tau)\, F(u),$$

$$F(u) = \mathrm{sn}(u, k)\,\mathrm{cn}(u, k)\,\mathrm{dn}(u, k), \qquad u = \omega(\tau + \tau_0). \tag{4.1}$$

Consequently, with $T_k = \frac{m}{n}T_0$, the generating function (3.2) in this case is given by

$$M_k(\tau_0) = 4k^2\omega \int_0^{nT_k} Z(\tau)F\big(\omega(\tau + \tau_0)\big)\, d\tau. \tag{4.2}$$

5. EXISTENCE OF SUBHARMONIC SOLUTIONS

In order to guarantee the existence of periodic solutions of the perturbed system (2.6) with period $nT_k = mT_0$, we must prove that the function $M_k(\tau_0)$ has a simple zero. We have computed this function in the case of a harmonic oscillation of the suspension point, both vertical and horizontal oscillations, and found it to be identically zero. However, for the anharmonic oscillator it is nonzero as we will see in this section.

To compute the integral in (4.2), we use the following Fourier series expansions for the functions $\mathrm{sn}^2 u$, $\mathrm{cn}^2 u$, $\mathrm{dn}^2 u$, see [12] (for more information on elliptic functions see also [2, 4] or [9])

$$\mathrm{sn}^2 v = S - b\sum_{r=1}^{\infty} Q_r \cos\left(\frac{r\pi}{K}v\right), \quad \mathrm{cn}^2 v = C + b\sum_{s=1}^{\infty} Q_s \cos\left(\frac{s\pi}{K}v\right), \quad \mathrm{dn}^2 v = D + k^2 b\sum_{t=1}^{\infty} Q_t \cos\left(\frac{t\pi}{K}v\right), \tag{5.1}$$

where $Q_r = \dfrac{rq^r}{1-q^{2r}}$, $q = \mathrm{e}^{-\frac{\pi K'}{K}}$, $b = \dfrac{2\pi^2}{K(k)^2 k^2}$ and

$$S = \frac{K(k) - E(k)}{K(k)k^2}, \quad C = \frac{E(k) - K(k)k'^2}{K(k)k^2}, \quad D = \frac{E(k)}{K(k)}, \tag{5.2}$$

with $k' = \sqrt{1-k^2}$ and $K' = K(k')$. Do not confuse the positive integer t here with time t.

Using the expressions of sn^2, cn^2 and dn^2 with k_0 and τ in place of k and v, we compute from (2.4) the following expression for $Z(\tau) = \Sigma(\sigma)$, $\sigma = \frac{K_0}{\pi}\tau$,

$$Z(\tau) = (CD - SD - k_0^2 SC) + U\sum_{p=1}^{\infty} Q_p \cos(p\tau) + V\sum_{r,s=1}^{\infty} Q_r Q_s \cos(r\tau)\cos(s\tau), \tag{5.3}$$

where

$$U = 2(Ck_0^2 - Sk_0^2 + D)b_0 \quad \text{and} \quad V = 3k_0^2 b_0^2. \tag{5.4}$$

In the proof of the proposition below we use the following lemma.

Lemma 1. *Let the positive integers m and n be relatively prime and assume that m is odd. Let the positive integers d and t satisfy the equality $md = 2nt$. Then $d = 2nl$ and $t = ml$ for some l.*

Proof. By hypothesis $md = 2nt$, so n divides md, hence it divides d, that is, $d = n\tilde{l}$ and therefore $m\tilde{l} = 2t$. Since m is odd, $\tilde{l}$ is even, say $\tilde{l} = 2l$. So $t = ml$ and $d = 2nl$. □

Proposition 1. *Assume that m is odd. Then the function* (4.2) *has the expression*

$$M_k(\tau_0) = \frac{2k^2 b m^2 \pi^2 \omega}{K}\sum_{l=1}^{\infty}\big(UA_l + VB_l\big)\sin(2nl\tau_0), \tag{5.5}$$

where U and V are given in (5.4) *and*

$$A_l = lQ_{2nl}Q_{ml} \quad \textit{and} \quad B_l = \sum_{r=1}^{\infty} lQ_{2nl+r}Q_r Q_{ml}. \tag{5.6}$$

Proof. From (4.1) we have $F(u) = \frac{1}{2}\frac{d}{du}\mathrm{sn}^2 u$ and since $\mathrm{sn}^2 u = S - b\sum_{r=1}^{\infty} Q_r \cos\left(\frac{r\pi}{K}u\right)$, we get

$$F\big(\omega(\tau+\tau_0)\big) = \frac{b\pi}{2K}\sum_{t=1}^{\infty} tQ_t\Big[\sin\left(\frac{t\pi\omega}{K}\tau_0\right)\cos\left(\frac{t\pi\omega}{K}\tau\right) + \cos\left(\frac{t\pi\omega}{K}\tau_0\right)\sin\left(\frac{t\pi\omega}{K}\tau\right)\Big].$$

It follows from (5.3) that the integral (4.2) is given by

$$M_k(\tau_0) = \frac{2k^2 b\pi\omega}{K}\sum_{t=1}^{\infty} tQ_t \sin\left(\frac{t\pi\omega}{K}\tau_0\right)\int_0^{2m\pi} Z(\tau)\cos\left(\frac{t\pi\omega}{K}\tau\right)d\tau. \tag{5.7}$$

By (5.3) and (5.7) we have to compute the integrals

$$J_{p,t} = \int_0^{2m\pi} \cos(p\tau)\cos\Big(\frac{t\pi\omega}{K}\tau\Big)\,d\tau \quad \text{and} \quad J_{r,s,t} = \int_0^{2m\pi} \cos(r\tau)\cos(s\tau)\cos\Big(\frac{t\pi\omega}{K}\tau\Big)\,d\tau.$$

Using the trigonometric identity

$$\cos(p\tau)\cos\Big(\frac{t\pi\omega}{K}\tau\Big) = \frac{1}{2}\Big[\cos\Big(p+\frac{t\pi\omega}{K}\Big)\tau + \cos\Big(p-\frac{t\pi\omega}{K}\Big)\tau\Big],$$

and noticing that because $T_k = \frac{m}{n}T_0$ and $T_0 = 2\pi$, we have

$$\Big(p \pm \frac{t\pi\omega}{K}\Big)nT_k = pmT_0 \pm 4nt\pi = (mp \pm 2nt)2\pi,$$

we get $J_{p,t} = m\pi$ if $mp - 2nt = 0$ and $J_{p,t} = 0$ if $mp - 2nt \neq 0$.

Analogously, the two identities coming from

$$\cos(r\tau)\cos(s\tau)\cos\Big(\frac{t\pi\omega}{K}\tau\Big) = \frac{1}{2}\big[\cos(r+s)\tau) + \cos(r-s)\tau\big]\cos\Big(\frac{t\pi\omega}{K}\tau\Big)$$

will lead to $J_{r,s,t} = \frac{1}{2}m\pi$ if $m(r+s) - 2nt = 0$ and also if $m(r-s) - 2nt = 0$ with $r > s$ and $m(r-s) + 2nt = 0$ with $r < s$, while $J_{r,s,t} = 0$ for any other values of the positive integers r, s, t.

From (5.3) and (5.7) we get the expression

$$M_k(\tau_0) = \frac{2k^2 b\pi\omega}{K}\sum_{t=1}^{\infty} tQ_t \sin\Big(\frac{t\pi\omega}{K}\tau_0\Big)\Big[U\sum_{p=1}^{\infty} Q_p J_{p,t} + V\sum_{r,s=1}^{\infty} Q_r Q_s J_{r,s,t}\Big].$$

Using the above considerations about $J_{p,t}$ and $J_{r,s,t}$ and Lemma 1 gives the expression (5.5) of $M_k(\tau_0)$ since $\frac{ml\pi\omega}{K} = 2nl$. □

We now use the lemma below to prove that the series $\sum_{l=1}^{\infty} A_l$ and $\sum_{l=1}^{\infty} B_l$ are convergent and, in fact, define analytic functions of the module k.

Lemma 2. *Let* $0 < \beta < 1$ *and* $Q_r = \dfrac{r\beta^r}{1-\beta^{2r}}$, $\quad r = 1, 2, 3, \ldots$. *We have the following facts:*

1. *The series* $\sum_{r=1}^{\infty} Q_r$ *and* $\sum_{r=1}^{\infty} rQ_r$ *converge;*
2. *The sequences* (Q_r) *and* (rQ_r) *are ultimately decreasing and their limits are zero;*
3. *For any subsequence* r_i *of* $r = 1, 2, 3, \ldots$, *the series* $\sum_{r_i=1}^{\infty} Q_{r_i}$ *converges;*
4. *If for the sequence* (a_r) *the partial sums* $A_n = \sum_{r=1}^{n} a_r$ *are bounded and* $b_1 \geqslant b_2 \geqslant b_3 \geqslant \ldots$ *with* $b_i \to 0$ *as* $i \to \infty$, *then the series* $\sum_{r=1}^{n} a_r b_r$ *converges.*

Proof. (1) The series $\sum_{r=1}^{\infty} rQ_r$ converges by the ratio test, since as $r \to \infty$,

$$\frac{(r+1)Q_{r+1}}{rQ_r} = \Big(\frac{r+1}{r}\Big)^2 \beta\,\frac{1-\beta^{2(r+1)}}{1-\beta^{2r}} \to \beta < 1.$$

(2) The function $\phi(x) = \dfrac{x^2\beta^x}{1-\beta^{2x}}$ is ultimately decreasing, since

$$\phi'(x) = \frac{x\beta^x(2 - 2\beta^{2x} + x\log\beta)}{(1-\beta^{2x})^2} < 0, \quad \text{for large } \ x.$$

So, the sequence (rQ_r) is ultimately decreasing and, by (1), the limit of rQ_r is zero.
(3) Let $c_{r_i} = Q_{r_i}$ and $c_r = 0$ for $r \neq r_i$. Then $c_r \leqslant Q_r$ for all r, so $\sum_{r=1}^{n} c_r$ converges.
(4) See [7], page 61. □

To show the analyticity of the functions defined by the series $\sum_{l=1}^{\infty} A_l$ and $\sum_{l=1}^{\infty} B_l$ as functions of k, we will use complex function theory, so we have to extend these functions to a neighborhood of the real axis, inside the unit disk.

The power series of $K(k)$ and $E(k)$ are given by (see, for instance, [4])

$$K(k) = \frac{\pi}{2}\Big\{1 + \sum_{r=1}^{\infty}\Big[\frac{(2r-1)!!}{(2r)!!}\Big]^2 k^{2r}\Big\}, \quad E(k) = \frac{\pi}{2}\Big\{1 - \sum_{r=1}^{\infty}\Big[\frac{(2r-1)!!}{(2r)!!}\Big]^2 \frac{k^{2r}}{2r-1}\Big\}, \tag{5.8}$$

where $r!! = r(r-2)(r-4)\cdots$, and they are convergent in the unit disk, $D : |z| < 1$, by the ratio test. Therefore, $K(z)$ and $E(z)$ are analytic functions of z in the disk D.

Therefore, $q(z) = \mathrm{e}^{-\frac{K'(z)}{K(z)}}$ is analytic in a narrow band of the real axis inside D. Now $0 \leqslant q(k) < 1$ for $0 \leqslant k < 1$, see Markeev [4] and the figure on the right. Fix any positive number $\delta < 1$ and let $\beta = 1 - \delta$ and $q(\beta) = 1 - \alpha$. Since $q(z)$ is continuous in D for each k in the interval $[0, \beta]$ there is a small rectangle $R_k = k - \epsilon_k, k + \epsilon_k] \times [-h_k, h_k]$ on which we have $q(z) \leqslant 1 - \frac{\alpha}{2}$. As the interval $[0, \beta]$ is compact, we can cover it by a finite number of these rectangles, and taking the smallest of their thickness, we get a strip S: $[0, \beta] \times [-h, h]$ on which the function $q(z)$ is bounded by $1 - \alpha/2$.

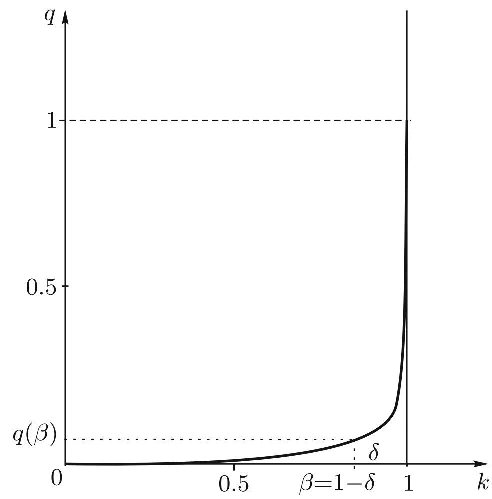

Fig. 4. Graph of the function $q(k)$.

Therefore, for all z in this strip we have

$$|Q_r(z)| = \left|\frac{rq(z)^r}{1 - q(z)^{2r}}\right| \leqslant \frac{rq(\beta)^r}{1 - q(\beta)^{2r}} = Q_r(\beta),$$

and $Q_r(z)$ is a sequence of analytic functions in an open neighborhood of the strip S.

Since by Lemma 2 the numerical series $\sum Q_r(\beta)$ converges, the series $\sum Q_r(z)$ defines an analytic function on S.

Similarly, the convergence of the series $\sum_{l=1}^{\infty} lQ_{2nl}(\beta)Q_{nl}(\beta)$ implies the uniform convergence of the series of analytic functions $\sum_{l=1}^{\infty} lQ_{2nl}(z)Q_{nl}(z)$, so its sum is an analytic function on S. In the same way we prove the analyticity of the function defined by the series

$$\sum_{l=1}^{\infty}\sum_{r=1}^{\infty} lQ_{2nl+r}(z)Q_r(z)Q_{ml}(z).$$

The analyticity of these functions on the strip S gives the analyticity of the real functions of k defined by their restrictions to the interval $[0, \beta]$.

Using now the theorem on sequences of differentiable functions on a closed interval (see Theorem 7.17 in Rudin [7]), we get as a corollary to Proposition 1 the following result.

Proposition 2. *For the derivative of M_k at $\tau_0^* = 0$ we have the expressions*

$$\frac{d}{d\tau_0} M_k(\tau_0^*) = \frac{4k^2 bm^2 n\pi^2 \omega}{K} \big(A^* U + B^* V\big), \tag{5.9}$$

where A^* and B^* are analytic functions of k on the closed interval $[0, \beta]$ and U, V are the functions of k_0 given in (5.4).

Proof. Differentiate the sequence of partial sums of (5.5) and use the above-mentioned theorem in [7]. For the terms in $A^* = \sum_{l=1}^{\infty} A_l^*$ and $B^* = \sum_{l=1}^{\infty} B_l^*$ we get the expression

$$A_l^* = l^2 Q_{2nl} Q_{ml} \quad \text{and} \quad B_l^* = \sum_{r=1}^{\infty} l^2 Q_{2nl+r} Q_r Q_{ml}.$$

□

Proof (*of Theorem* 1). The coefficients $A^*(k)$ and $B^*(k)$ in (5.9) are obviously positive for all k in the interval $[0, \beta]$. Clearly, $V = 3k_0^2 b_0^2$ is a positive function of k_0 for all k_0 in the interval $[0, 1)$. Let us study the sign of U. Using the expressions of C, D, S in (5.2), we have for U in (5.4)

$$U = 2\big((C - S)k_0^2 + D\big)b_0 = 2b_0 \left[\frac{2(E - K)}{Kk^2} k_0^2 + k_0^2 + \frac{E}{K}\right]. \tag{5.10}$$

Using (5.8) we check that $\frac{2(E-K)}{Kk^2}$ is a bounded function of k in $[0, \beta]$. Since $\frac{E}{K}$ is continuous and positive on $[0, 1)$, it is bounded from below by a positive number on the compact interval $[0, \beta]$. Hence, there exists a $k_0^* > 0$ sufficiently small such that $U(k_0) > 0$ for all $k_0 \leqslant k_0^*$.

Since ϵ goes to zero with k_0 corresponding to k_0^*, there is an $\epsilon_n > 0$ such that for all $\epsilon \leqslant \epsilon_n$ and all $k \in [0, \beta]$ the right-hand side of (5.9) is positive, that is, the derivative of the Melnikov function at $\tau_0^* = 0$ is positive. So this function has a simple zero and therefore the system (2.6) has a periodic solution of period $nT_k = mT_0$. □

6. CONCLUDING REMARKS

1. The oddness of m in Proposition 1 is not essential, it only simplifies a little bit the expressions in the integration of (5.7).

2. Theorem 1 gives a kind of density of periodic orbits in the phase space of the unperturbed pendulum. Indeed, since the period T_k is an increasing function of k, by the density of the rational numbers we can find, for any k, rational multiples of T_0 arbitrarily close to T_k.

3. As k approaches 1, ϵ_n tends to zero because the quotient E/K goes to zero, hence its lower bound on $[0, \beta]$ becomes small, so k_0, and with it ϵ_n, goes to zero.

4. The question remains whether subharmonic solutions of the perturbed pendulum exist for harmonic oscillations of the point of suspension. The proof of this existence requires a different technique because we have computed the generating function and found it to vanish identically in this case. We note, however, that in a small neighborhood of the stable equilibrium subharmonic solutions for (horizontal) harmonic oscillations of the point of suspension do exist, as was proved in [13].

5. It is not yet clear what the anharmonic perturbations lead to in comparison with the harmonic perturbations, but there must be differences as the previous item suggests. Maybe one should explore more the asymmetry of the anharmonic perturbations, see Fig. 2.

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